

PARITOSH KIRAN GANDRE | Machine Learning Engineer

+1 330-554-8445 | paritoshkrcg@gmail.com | GitHub | LinkedIn

Professional Summary

Machine Learning Engineer with 2+ years building production real-time systems. Currently leading ML architecture for sub-second latency voice translation platform. Expertise in inference optimization, open-source ML frameworks, and stress-testing systems at scale.

Technical Skills

Languages: Python, SQL, R

ML Frameworks & Libraries: TensorFlow, PyTorch, scikit-learn, Keras, CNNs, NLP (spaCy, NLTK), ARIMA, SHAP, XGBoost, NLLB-200, CTranslate2

Deep Learning & NLP: Convolutional Neural Networks, NLP pipelines, sentiment analysis, recommendation systems, LLM prompt engineering

Cloud & Infrastructure: AWS (EC2, S3, Lambda), Docker, Apache Airflow, CI/CD pipelines, REST APIs, ETL pipelines, NVIDIA infrastructure

Data Analysis & Visualization: Matplotlib, Seaborn, Power BI, Tableau, A/B testing, hypothesis testing, statistical modeling

Big Data & Platforms: Spark (familiar), MS Fabric (familiar), SQL-based data pipelines

Software Practices: Version control (Git), hyperparameter tuning, model evaluation, cross-validation, documentation

Experience

Machine Learning Engineer

Nov 2025 – Present

OViiE AI | Remote

- Architecting end-to-end inference pipelines (ASR to neural machine translation to TTS) optimized for sub-second latency; leading technical decisions on inference infrastructure and model deployment strategies
- Designed and implemented stress-testing framework and observability systems to measure real-time ML performance at scale; established latency monitoring and bottleneck analysis practices that reduced mean inference time by 12% through targeted optimization
- Evaluating and selecting open-source models and frameworks (NLLB-200, CTranslate2, Piper) for cost-performance tradeoffs across multiple language pairs and NVIDIA GPU targets
- Leading Phase 2 self-hosted architecture design for multi-tenant SaaS deployment, enabling scalable inference without vendor dependency

Research Assistant – Deep Learning for Genomic Data

Feb 2025 – Dec 2025

Kent State University | Kent, OH

- Designed CNN models classifying 40K+ genomic sequences with 97% accuracy
- Automated data preprocessing and training pipelines using Python/Docker, reducing manual handling by 80%

Data Scientist – AI Engineering

May 2024 – Dec 2024

Inke (SafeSpace) | Orlando, FL (Remote)

- Designed and deployed NLP models (sentiment analysis, recommendation systems) using spaCy and NLTK, serving 100K+ daily requests with 85% accuracy
- Built end-to-end ML pipelines on AWS EC2 with Apache Airflow CI/CD, improving deployment reliability by 30% and reducing API latency by 30%

Data Analyst – Supply Chain & Operations

Apr 2024 – May 2025

Kent State University Dining Services | Kent, OH

- Built ARIMA-based predictive models for demand forecasting, reducing over-purchasing by 15% (approximately \$10K savings)
- Developed automated Power BI dashboards and SQL pipelines, cutting reporting latency by 30%

Data Scientist

Jan 2022 – Jun 2023

S L Kulkarni Cyril Graphics Pvt. Ltd. | Mumbai, India (Remote)

- Developed predictive maintenance ML models reducing machine downtime by 20%; deployed real-time anomaly detection dashboards using Flask and SQL
- Engineered ETL pipelines integrating production, service, and machine data, improving data reliability by 30%

Key Projects

- **Genomic Sequence Classification (CNN-HMM Hybrid)** – CNN-HMM architecture classifying DNA motifs with 97% accuracy using TensorFlow
- **Pose Corrector App (Computer Vision)** – Cross-platform exercise form-correction app with sub-50ms mobile latency using MediaPipe and TensorFlow Lite

- **AI-Powered Content Generator (LLM Automation)** – Flask REST endpoint with GPT integration and scheduled job automation

Education

Kent State University, Kent, OH

Aug 2023 – May 2025

Master of Science, Data Science | GPA: 3.4/4.0

Relevant Coursework: Artificial Intelligence (TensorFlow, PyTorch, Keras), Statistical Learning, Applied & Computational Statistics

Certifications

- Fundamentals of Accelerated Computing with CUDA Python – NVIDIA

Mar 2026
- AWS Cloud Technical Essentials – Amazon Web Services

Oct 2025
- Data Analysis with R Programming – Google

Nov 2024